# Diversity of baby names through history

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# Data Set

## Baby Names from Social Security Card Applications - National Data

Records of baby names, gender, and frequency by year.

- **Source:**
  - Social Security Administration
- **How collected:**
  - From Social Security card applications
- **Why collected:**
  - Track population + naming trends
- **Useful variables:**
  - Name
  - Year
  - Gender
  - Count (# of babies with that name)
- **URL:**
  - https://catalog.data.gov/dataset/baby-names-from-social-security-card-applications-national-data?from_hint=eyJzb3J0IjoicG9wdWxhcml0eSJ9

# Real-World Application

**We act as baby naming consultants helping couples choose baby names based on trends, uniqueness, and future popularity.**

**Couples want a name that is:**

- unique
- Conventional for the time
- Enduring and timeless

Q1

# How do individual baby names differ in their rise, peak, and decline patterns over time?

- **Why it is interesting:**
  - Shows cultural trends
  - Shows the rise in popularity and eventually decline
  - Over time could show waves of popular names
- **Why it is inferential:**
  - Looking at patterns, not exact values
  - Shows if a name you like is on an increase or down turn
- **Variables:**
  - Year vs Name Count

Q2

# How has the distribution of baby names changed over time?

- **Why interesting:**
  - Cultural shift toward individuality
  - Cultural shift toward idols or popular figures
- **Why inferential:**
  - Measuring spread/diversity
- **Variables:**
  - Year vs number of unique names

Q3

# What does past trend data suggest about a name's future popularity?

- **Why interesting:**
  - Helps predict future popularity
  - What generations do returning names come from
- **Why inferential:**
  - Uses past trends to estimate future direction
- **Variables:**
  - Year vs Name Count
  - Year-to-year change (growth rate)

Q4

# How do popularity trends differ between male and female names?

- **Why interesting:**
  - Tests stability vs volatility
  - Can show the difference in boy names to girl names (eg. buck vs lilly)
  - Can show the popularity of naming boys girls names and vice versa
- **Why inferential:**
  - Comparing two groups over time
- **Variables:**
  - Gender vs Name frequency trends

Q5

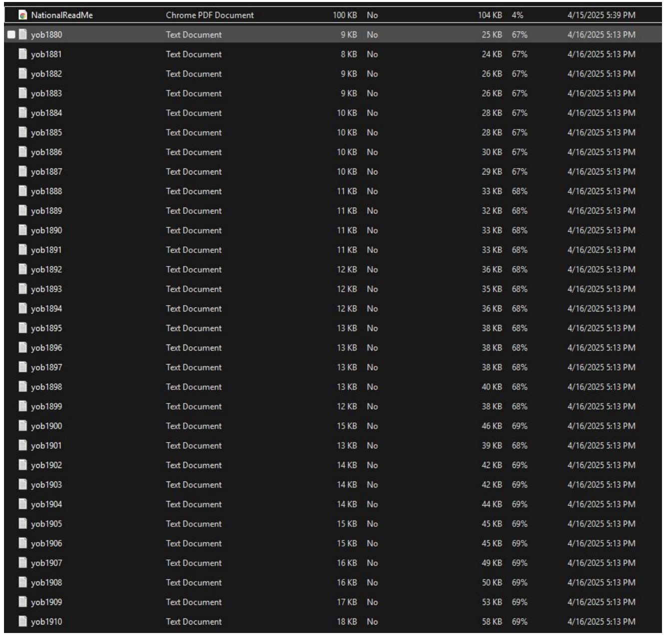
# What patterns show when names decline and later return in popularity?

- **Why interesting:**
  - Cyclical trends
- **Why inferential:**
  - Identifying patterns over long time
- **Variables:**
  - Name vs Year vs Count

# Data and Question Alignment

### Dataset breakdown:

- Data is split into yearly files (yobYYYY.txt), which contain:
  - Name
  - Sex (M/F)
  - Number of occurrences
- Files are sorted by popularity
- Only includes names with ≥5 occurrences
- Time range: 1880 to present



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# Data and Question Alignment - Table

| Question | Data Used | Can Answer? |
|---|---|---|
| How do individual baby names differ in their rise, peak, and decline patterns over time? | Name, year, number/count | Yes |
| How do popularity trends differ between male and female names? | Sex, name, year, number/count | Yes |
| How has the distribution of baby names changed over time? | Year, total births, top 10 name counts, number of unique names | Yes |
| What does past trend data suggest about a name’s future popularity? | Name, year, number/count, year-to-year change, trend slope | Yes |
| What patterns show when names decline and later return in popularity? | Name, year, number/count, gaps in popularity, multiple peaks | Yes |

# Most Promising/Interesting Questions

1. How do individual baby names differ in their rise, peak, and decline patterns over time?
2. How has the distribution of baby names changed over time?
3. What does past trend data suggest about a name's future popularity?

# How do individual baby names differ in their rise, peak, and decline patterns over time?

## Metrics:

- Yearly name count
- Year-to-year change
- Percent growth rate
- Peak year and peak count

## Plots / Visualization:

- Line graph (Year vs Name Count)
- Smoothed trend line (moving average)
- Highlight peak points

## Tools / Python Packages:

- Pandas
- Numpy
- Matplotlib

# How has the distribution of baby names changed over time?

## Metrics:

- Top 10 Share
  - births with top 10 names / total births
- Number of unique names per year
- Total births per year

## Plots / Visualization:

- Line graph: Year vs Top 10 Share
- Line graph: Year vs Number of Unique Names

## Tools / Python Packages:

- Pandas
- Numpy
- Matplotlib

# What does past trend data suggest about a name’s future popularity?

**Metrics:**

- Recent growth rate
- Moving average
- Trend slope
- Predicted direction: increasing, decreasing, or stable

**Plots / Visualization:**

- Line graph: Year vs Name Count
- Trend line showing projected direction

**Tools / Python Packages:**

- Pandas
- Numpy
- Matplotlib

# Rough Justification: What does past trend data suggest about a name's future popularity?

This graph shows Olivia's popularity as a percentage of total births over time. The moving average smooths the yearly changes, while the projected trend estimates future popularity based on recent patterns. I expect the name to remain popular, but the trend may begin to level off instead of continuing to rise sharply.

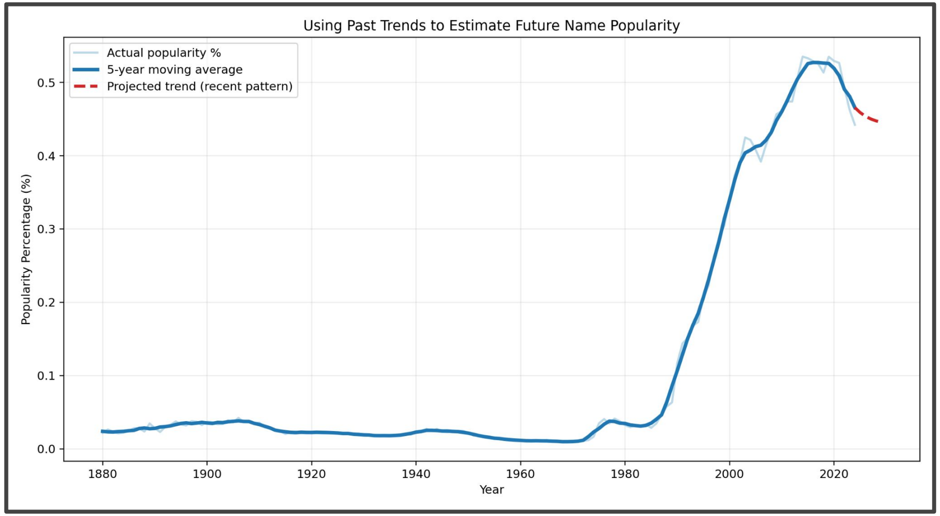
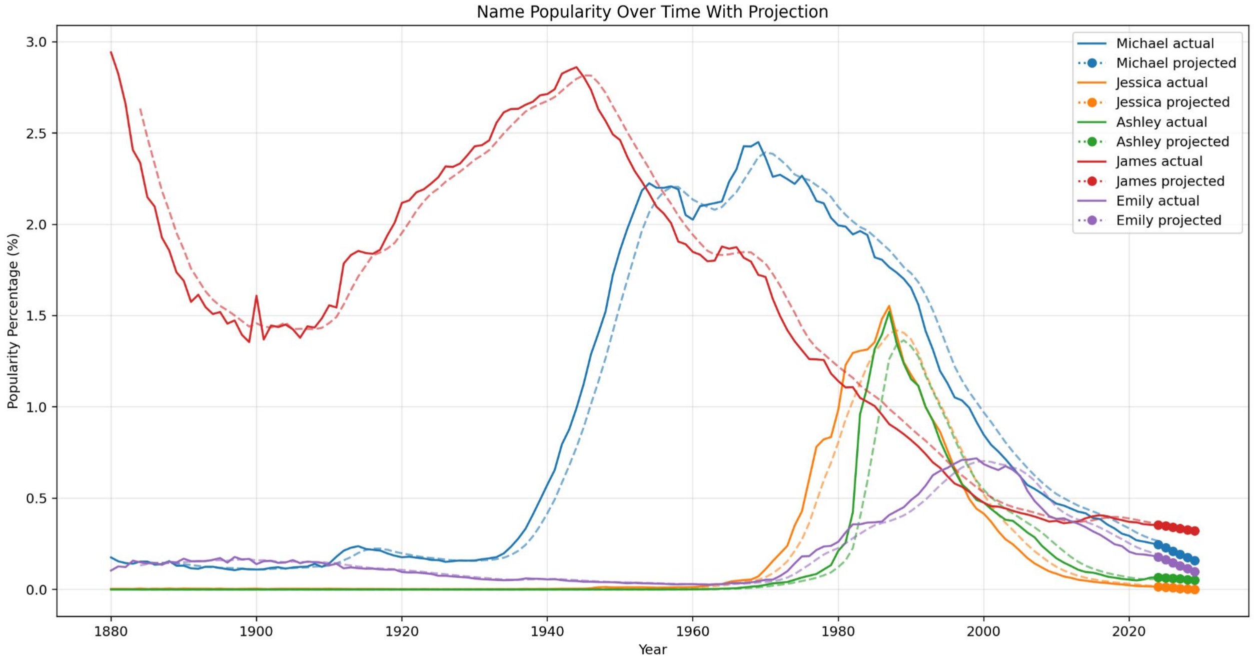



Figure 2: Future Prediction

# Sketches at Answers: What does past trend data suggest about a name's future popularity?



# Sketches at Answers: What does past trend data suggest about a name's future popularity?

```
Michael (M)
  count: 8,189  |  popularity: 0.2460%  |  growth: -19.55%
  slope: -0.017578  |  direction: decreasing
  projection: 2025=0.2284%  2026=0.2109%  2027=0.1933%  2028=0.1757%  2029=0.1581%

Jessica (F)
  count: 526  |  popularity: 0.0158%  |  growth: -43.39%
  slope: -0.003154  |  direction: decreasing
  projection: 2025=0.0126%  2026=0.0095%  2027=0.0063%  2028=0.0032%  2029=0.0000%

Ashley (F)
  count: 2,231  |  popularity: 0.0670%  |  growth: +14.60%
  slope: -0.002886  |  direction: decreasing
  projection: 2025=0.0641%  2026=0.0613%  2027=0.0584%  2028=0.0555%  2029=0.0526%

James (M)
  count: 11,793  |  popularity: 0.3543%  |  growth: -6.79%
  slope: -0.006543  |  direction: decreasing
  projection: 2025=0.3478%  2026=0.3412%  2027=0.3347%  2028=0.3281%  2029=0.3216%

Emily (F)
  count: 5,955  |  popularity: 0.1789%  |  growth: -24.78%
  slope: -0.016113  |  direction: decreasing
  projection: 2025=0.1628%  2026=0.1467%  2027=0.1306%  2028=0.1145%  2029=0.0983%
```

# Rough Justification: How has the distribution of baby names changed over time?

A decreasing line over time means names are becoming more diverse, fewer babies are being given the same top 10 names.

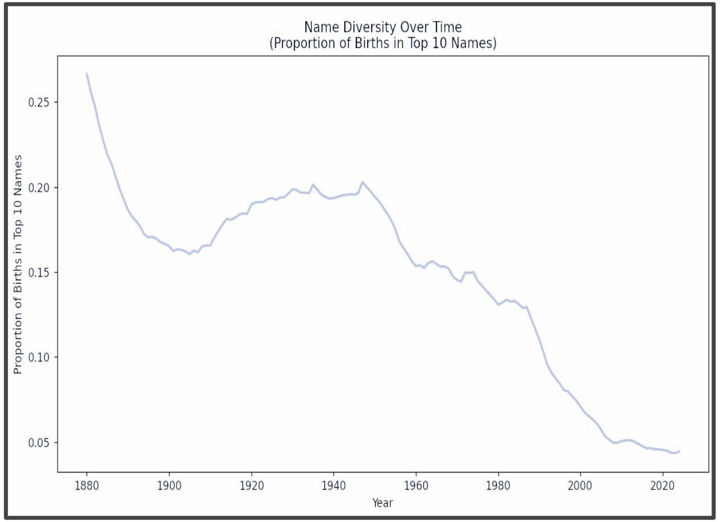



Figure 2: Name Diversity Graph

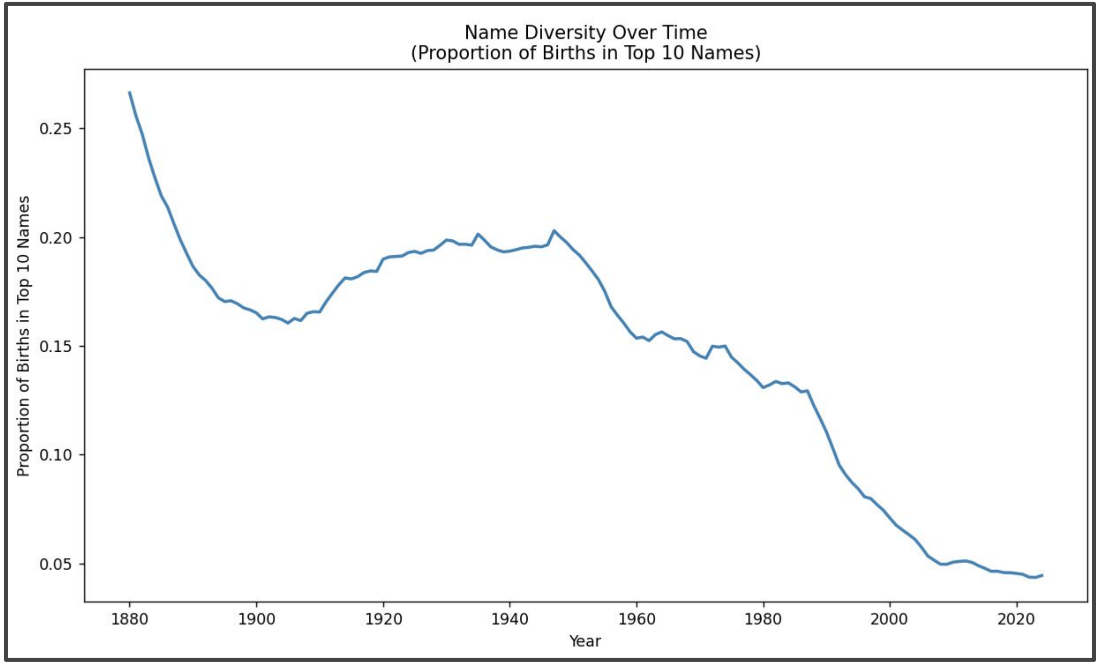
# Sketches at Answers: How has the distribution of baby names changed over time?

```
PS C:\Users\cmchi\OneDrive\baby names> & C:\Users\cmchi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/cmchi/OneDrive/
baby names/Name Diversity.py"
Least diverse year: 1880 — top 10 names accounted for 26.6% of births
Most diverse year:  2023 — top 10 names accounted for 4.3% of births

Proportion in 1880: 26.6%
Proportion in 2024: 4.4%
Overall change: -22.2 percentage points

Average top 10 share by decade:
  1880s: 22.6%
  1890s: 17.4%
  1900s: 16.3%
  1910s: 17.8%
  1920s: 19.3%
  1930s: 19.7%
  1940s: 19.7%
  1950s: 17.6%
  1960s: 15.3%
  1970s: 14.4%
  1980s: 12.9%
  1990s: 8.8%
  2000s: 5.9%
  2010s: 4.8%
  2020s: 4.4%
```

# Sketches at Answers: How has the distribution of baby names changed over time?



# Rough Justification: How do individual baby names differ in their rise, peak, and decline patterns over time?

For this example I took the 3 most popular female names in 1880 and created a rough estimate of how the popularity of their name changed over time.

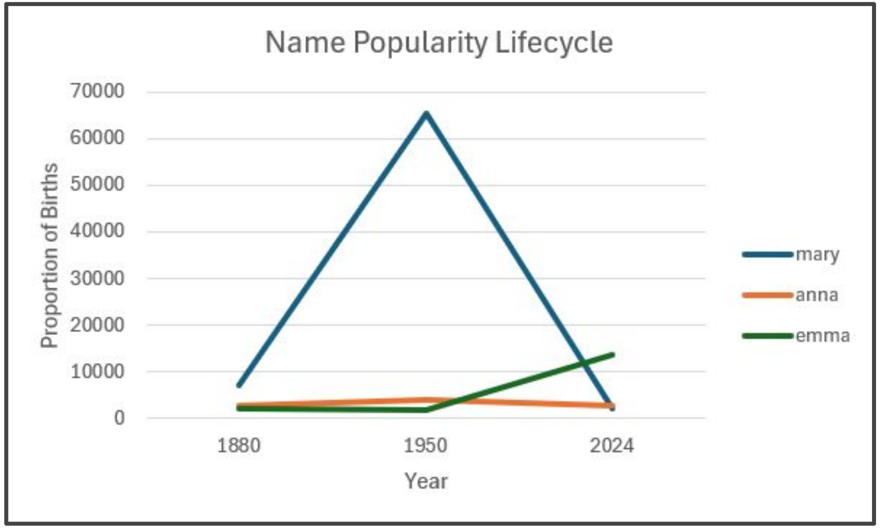
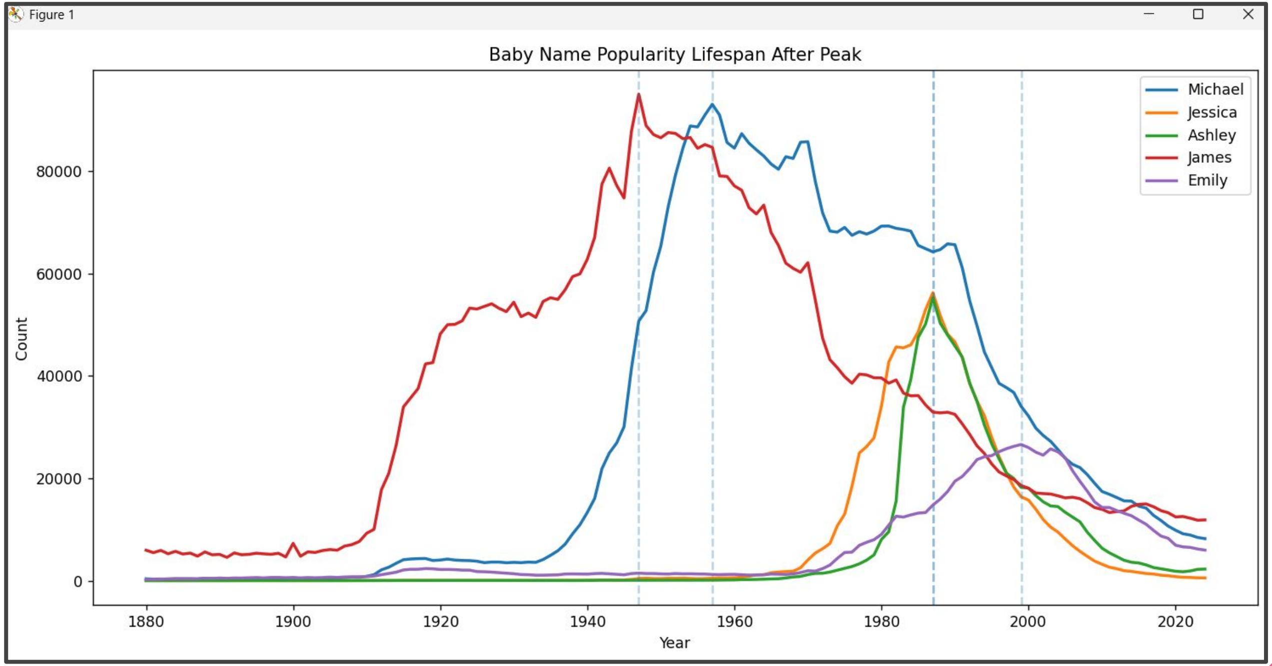



Figure 1: Example Name Popularity Graph

# Sketches at Answers: How do individual baby names differ in their rise, peak, and decline patterns over time?



# Sketches at Answers: How do individual baby names differ in their rise, peak, and decline patterns over time?

```
PS C:\Users\cmchi\OneDrive\baby names> & C:\Users\cmchi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/cmchi/OneDrive/baby names/new question.py"
```

| Name | Peak Year | Peak Count | 50% Decline Year | Years to 50% Decline | Decline Rate/Year |
|---|---|---|---|---|---|
| Michael | 1957 | 93041 | 1994 | 37 | -1386.7 |
| Jessica | 1987 | 56239 | 1995 | 8 | -1394.5 |
| Ashley | 1987 | 55262 | 1995 | 8 | -1324.1 |
| James | 1947 | 95020 | 1972 | 25 | -1142.5 |
| Emily | 1999 | 26587 | 2013 | 14 | -934.6 |

# Data Decisions: What does past trend data suggest about a name's future popularity?

**Answer found:** Yes. All five names analyzed show a decreasing trend. Michael is declining at -19.55% growth rate, Jessica at -43.39%, and James, the slowest, at -6.79%

**Expected relationship:** It was expected that past trends would continue into the future — the linear projections support this, though the model assumes a straight-line decline which may oversimplify reality

**Clear conclusion:** Yes — all analyzed names are projected to continue declining in popularity through 2029. James will decline the slowest due to its long cultural staying power, while Jessica is projected to nearly disappear by 2029

# Data Decisions: How has the distribution of baby names changed over time?

**Answer found:** Yes, very clearly. In 1880 the top 10 names accounted for 26.6% of all births. By 2023 that dropped to just 4.3% — a decline of 22.2 percentage points

**Expected relationship:** It was expected that names would become more diverse over time — this was strongly confirmed

**Clear conclusion:** Yes — name diversity has increased consistently every decade, with the sharpest drop occurring after the 1950s. Parents today choose from a far wider range of names than in previous generations

# Data Decisions:  How do individual baby names differ in their rise, peak, and decline patterns over time?

**Answer found:** Yes, names differ significantly in their life cycle patterns. Michael peaked in 1957 at 93,041 births and took 37 years to decline 50%, while Jessica and Ashley both peaked in 1987 and declined within just 8 years despite similar peak counts

**Expected relationship:** It was expected that higher peak counts would mean longer lifespans — this was partially confirmed by Michael and James, but Ashley and Jessica contradicted this

**Clear conclusion:** Yes — names that peaked in earlier decades (pre-1960s) tend to have much longer popularity lifespans than names that peaked in the 1980s-90s, suggesting cultural naming trends have accelerated over time

# Reflection of Process

Working with 145 real-world data files was more challenging than expected — issues like file paths, column names, and encoding problems had to be solved before any analysis could begin

The most surprising finding was that names peaking in the 1980s declined much faster than older names, which was not the initial expectation

This process showed that data rarely behaves exactly as expected — Ashley and Jessica had similar peak counts to Michael but completely different lifespans, proving that raw numbers alone don't tell the whole story

If starting over, more names across different eras would be included to make the conclusions more generalizable, and a non-linear model might better capture the natural rise and fall of name trends

# Reflection of Personal Growth

At the start of this course, working with real datasets and writing Python scripts felt overwhelming. Debugging file path errors and fixing column name mismatches were frustrating at first but built real problem-solving skills

The biggest growth was learning to not just write code that runs, but to think critically about what the results actually mean and whether they answer the original question

Study habits improved by breaking large problems into smaller steps rather than trying to solve everything at once

The baby names dataset was engaging because it connected to real cultural patterns, making the analysis feel meaningful rather than abstract